# Expanding Horizons: Single-Cell Multiomics Analysis on Galaxy

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## Introduction

Single-cell transcriptomics analysis has become standard practice on Galaxy, thanks to a number of tools and workflows. In our recent efforts, we expanded the tool suites from single-cell transcriptomics to epigenomics and multiomics, enabling researchers to integrate and cluster these modalities together within Galaxy. For single-cell ATAC-seq data analysis, we incorporated scalabele, Python-based SnapATAC2 and EpiScanpy packages into Galaxy. Furthermore, the underlying data format Anndata, facilitates exceptional interoperability with the well-established Galaxy Scanpy tool suite. Multimodal integration is performed through matrix factorization techniques from muon framework.

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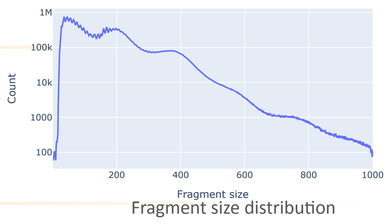
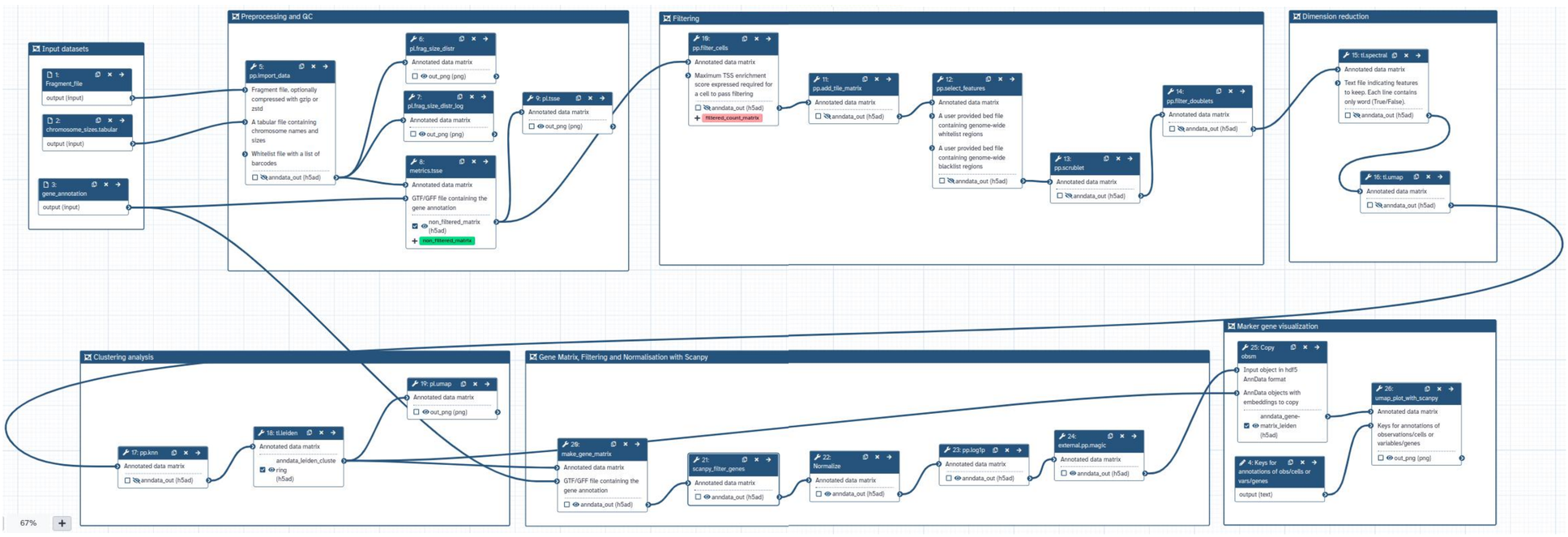
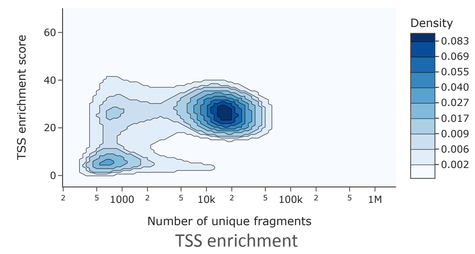
## Single-cell ATAC-seq standard workflow using SnapATAC2

- QC: Fragment size distibution
- QC: TSS enrichment
- Feature selection
- Doublet removal using Scrublet
- Spectral embedding for dimensionality reduction
- Gene activity matrix creation and normalization
- Imputation and data smoothing with MAGIC algorithm
- Clustering with Leiden, K-means and DBSCAN
- Marker gene identification and cluster annotation

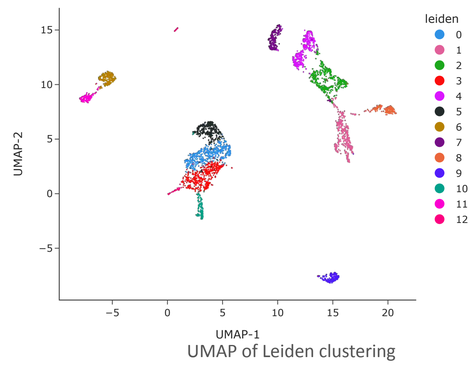


Fragment size distribution



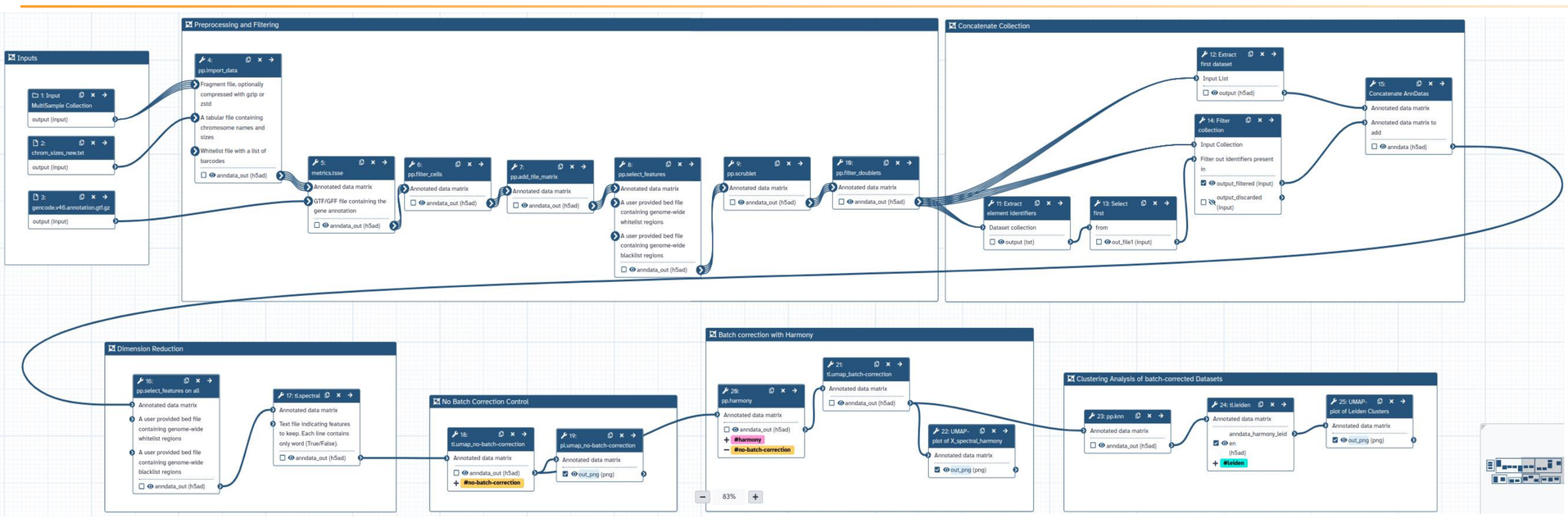
TSS enrichment



UMAP of Leiden clustering

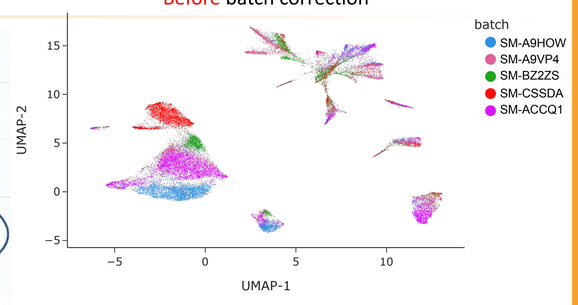
## Multi-sample workflow with batch correction

- Allows more than 2 batches
- Separate preprocessing of each batch
- Automated concatenation of batch Anndata objects
- Various batch correction algorithms: Harmony, MNN-Correct or Scanorama

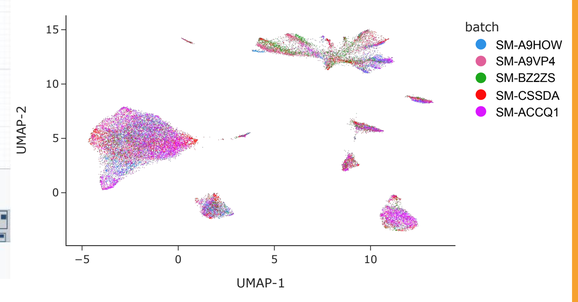



Workflow tested on 42k cells of human colon from 5 donors

Before batch correction

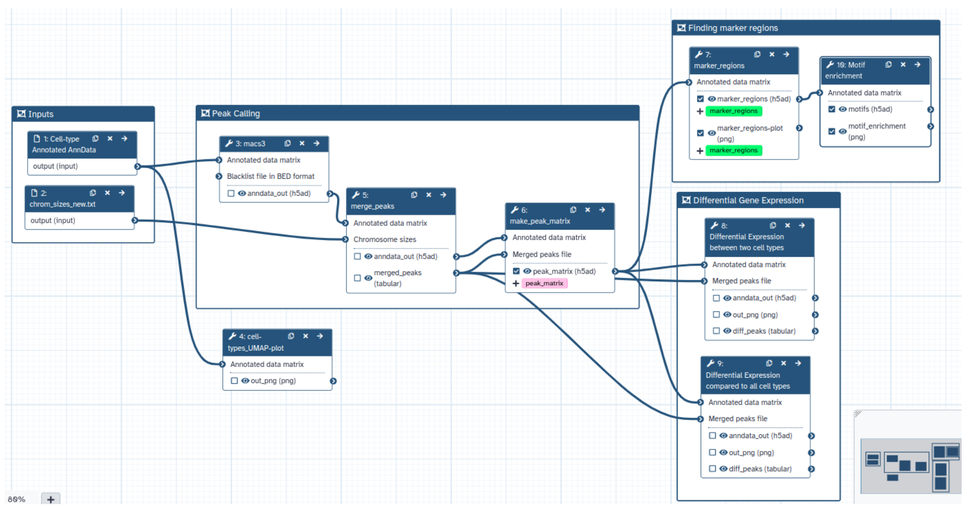



After batch correction

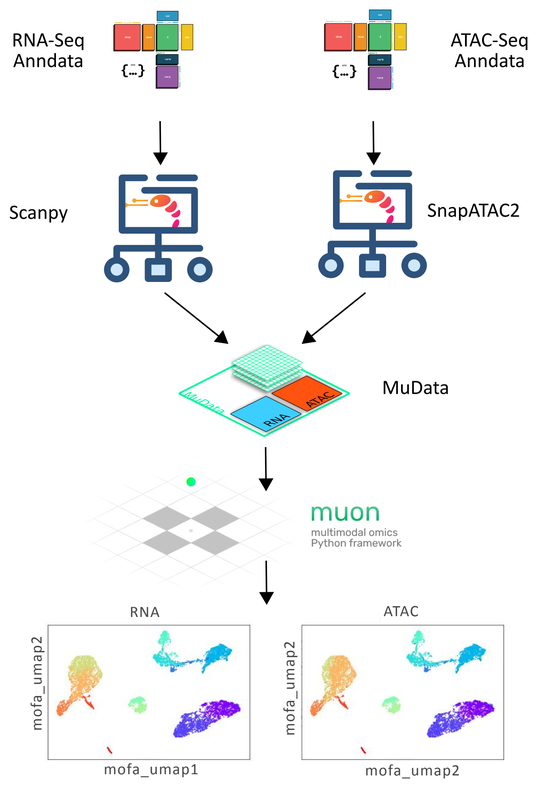



## Identify differentially accessible regions

- Cluster-level peak calling using macs3 and peak matrix generation
- Allows blacklist or whitelist to specifically select relevant features
- Find marker regions based on enriched peaks
- Motif-enrichment and regression-based differential test




## Multimodal integration using muon*

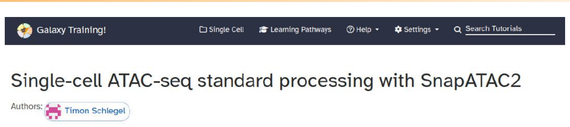



- Scanpy for scRANA-seq
- SnapATAC2 for scATAC-seq
- Combine both *Anndata* objects into *MuData*
- Easy extraction of Anndata objects for modality specific analysis
- Implements multi-omics factor analysis (MOFA)
- Anndata enables seamless interoperability among the tool suites

* muon Galaxy tool is under review

## Training material (WIP)

- Standard analysis
- Multi-sample batch correction
- Peak and motif analysis




## Funding and partners

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